Statistical Analysis and Validation Issues

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Introduction

- **Statistical analysis of data based on condition**
- **Validation issues**

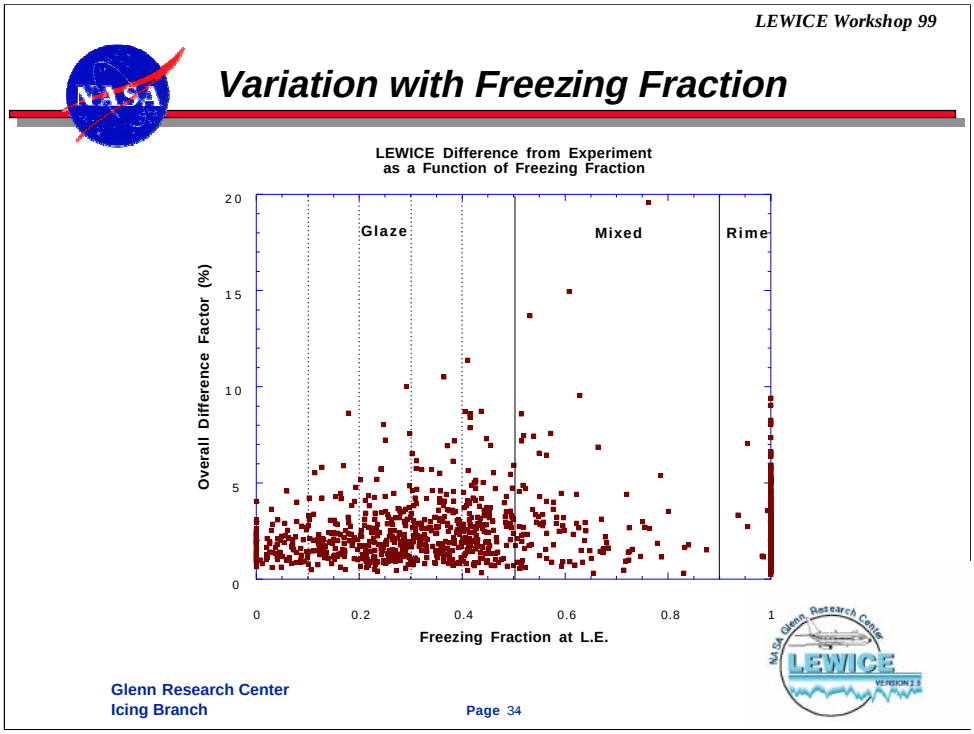
This presentation will cover a more detailed statistical look at the comparisons between LEWICE and experiment and also discuss some additional validation issues.

Variability as a Function of Condition

- What physical conditions cause the worst predictions in LEWICE?
- Answers could lead to further code improvements

Once the quantitative data has been compiled, it is easy to determine the relative errors from LEWICE as a function of the conditions run in the tunnel.

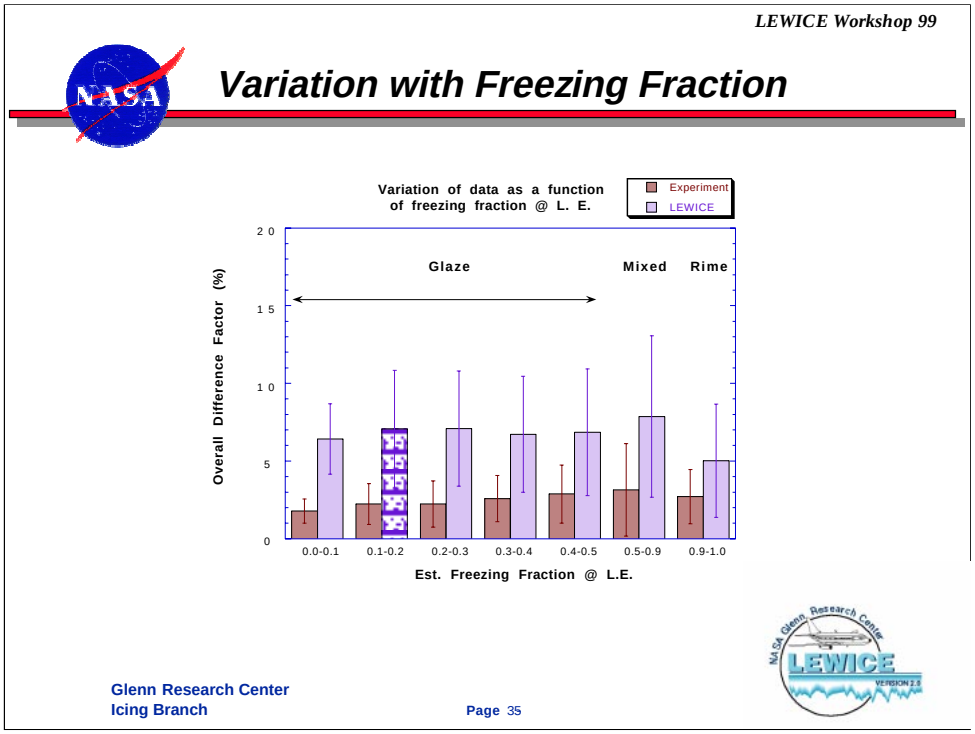
For each parameter, the data was divided into regions of different physical phenomena with some care taken to ensure that each region is represented by a statistically significant number of data points.



This plot shows the difference between LEWICE and each experimental ice shape as a function of the freezing fraction at the leading edge. In this particular figure, a scatter plot is shown to illustrate how the different regions were chosen. Each bin in the data has a similar number of data points, so the regions chosen illustrate that less data was taken in the freezing fraction region 0.5-0.9 than in other regions. The dividing lines between what is considered “glaze”, “mixed” and “rime” are somewhat arbitrary, but representative of those regions.

Additionally, freezing fraction is a calculated parameter, not a measured parameter. It was calculated from the experimental ice shape by taking the ration of the leading edge thickness to the maximum horn thickness. Qualitatively, this is representative. If the ice thickness at the leading edge is zero, obviously the freezing fraction is zero. In rime conditions, the maximum thickness is at the leading edge, so the ratio is one. In between these extremes, the ice shape moves from very pronounced horns to less visible ones, which is consistent with the “glaze” and “mixed” ice shape criteria.

To show detail, note that the y-axis has been expanded.

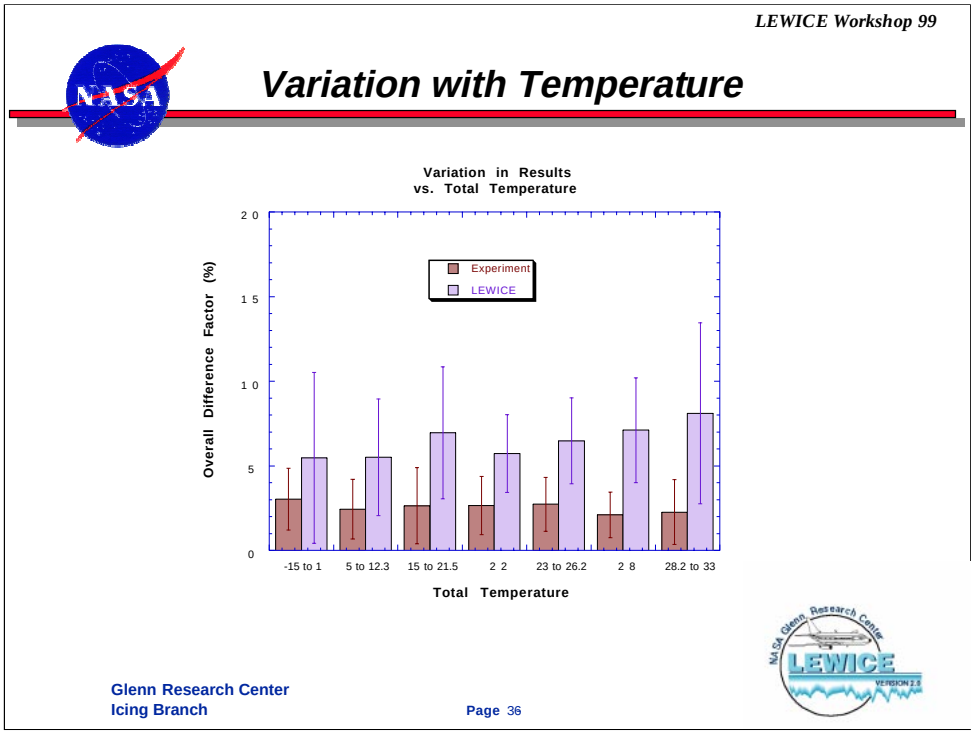


This plot shows the same information in a bar chart. The height of each bar represents the average for that bin while the error bar represents one standard deviation from the average for the data in that region. The previous plot showed only the LEWICE data for clarity, while this plot can show both the experimental variation and the LEWICE variation.

This plot shows that there is very little difference in the average error as a function of freezing fraction. While the error is slightly lower for a pure rime ice case, the difference from the glaze ice cases is less than 2%, well within the statistical variation of the results. The experimental error is even more consistent with freezing fraction, showing that the tunnel can accurately reproduce glaze ice shapes.

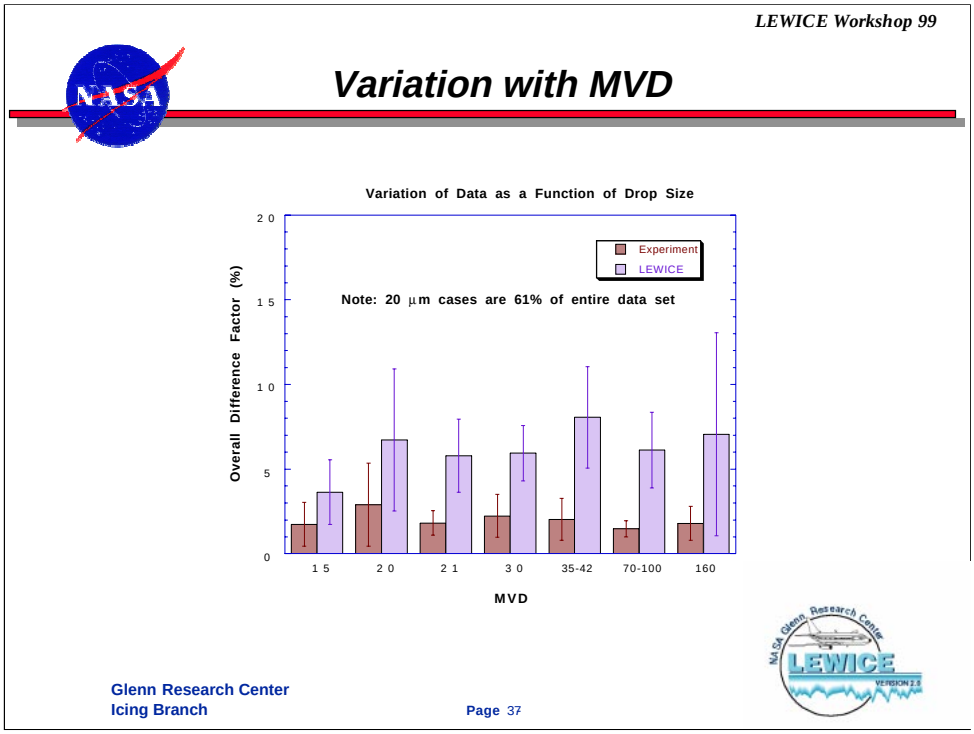
Conclusion: LEWICE 2.0 is very nearly as accurate in glaze or mixed icing conditions as in rime icing conditions.

Do you agree with this statement?



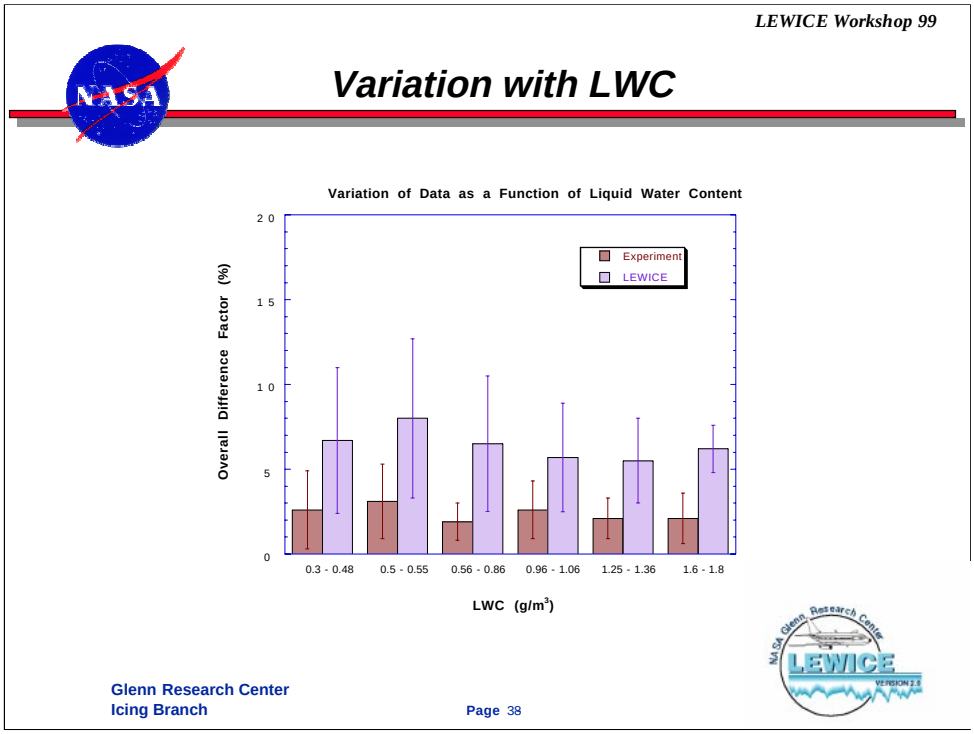
This plot shows the same data as the previous chart, except that the independent variable is the measured quantity of total temperature instead of freezing fraction. This plot shows a similar result as the previous chart. There is not a statistically significant difference between high temperature cases and low temperature cases. This chart also shows that careful attention needs to be taken in defining the ranges so that each bin size is well represented. In this chart, the largest bin size (the one with the most data points) is the 28˚F temperature bin. In each case, the bin ranges were chosen to include a statistically significant sample size while taking into account natural breaks in the data.

Conclusion: The variation of results with temperature is very weak to nonexistent.



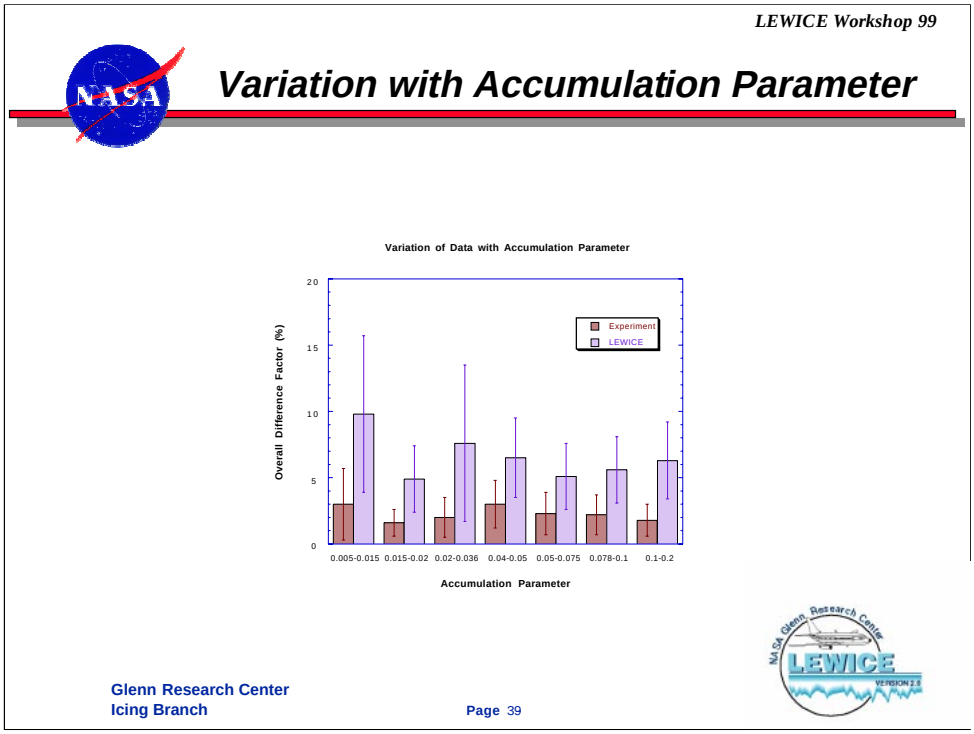
Continuing with the trend of the previous charts, this chart shows the variation as a function of median volume diameter drop size. In this case, the bins could not be equally divided as the 20 micron bin size encompasses 61% of the entire data set. Therefore, the other bins are underrepresented. This chart shows a smaller error for the 15 micron cases, but otherwise there is no significant difference in error even into the exceedence conditions. The large error bar for the last bin is an indication that the bin size is small.

Conclusion: Although more data is desired to conclusively prove this hypothesis, LEWICE 2.0 predicts ice shapes in exceedence conditions as accurately as it predicts ice shapes in nominal drop size conditions.



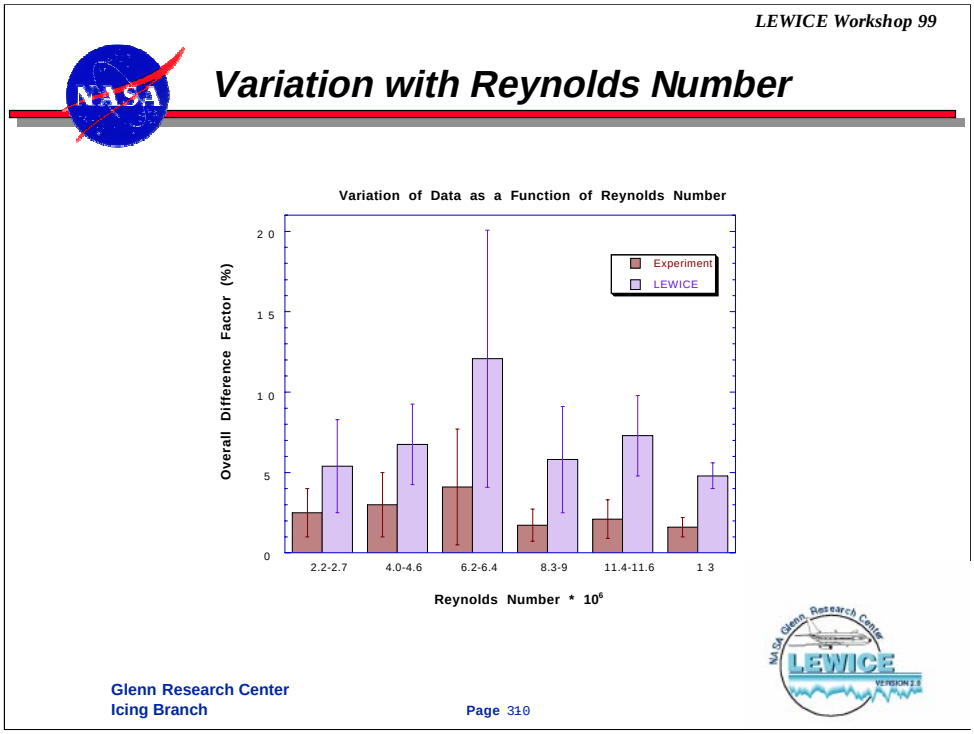
This plot shows the data as a function of Liquid Water Content. There are some higher error bars in low water content cases. This is understandable as the percent difference factors devised showed a clear tendency to magnify small absolute errors on small ice shapes as larger percentage errors. The ave3rage values do no show much of trend.

Conclusion: There do not appear to be any significant trends with respect to Liquid Water Content.

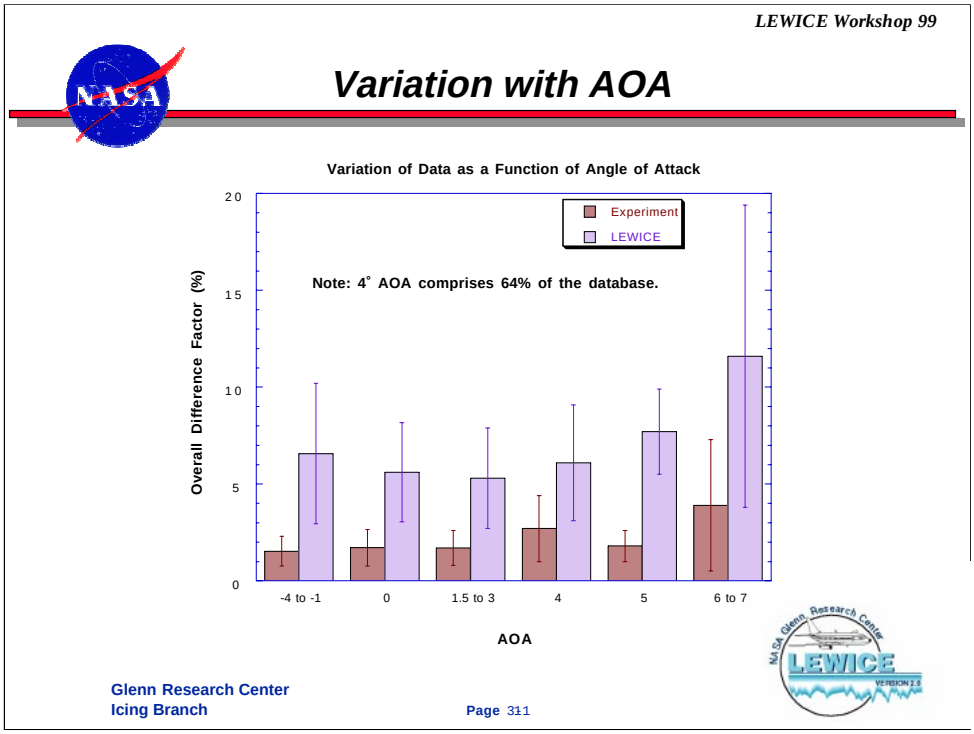


This chart shows the same data as the previous chart, but this time using Accumulation Parameter as the independent variable. In this case, there is a trend of higher error with decreasing Accumulation Parameter, except for the one bin which shows a much lower error than the other region. Since the sample size in this bin is considerable, it cannot be discounted. As a result, no conclusive trends are apparent.

Conclusion: There are no significant trends with Accumulation Parameter. If anything, the data shows smaller percentage errors as the ice shape grows larger.



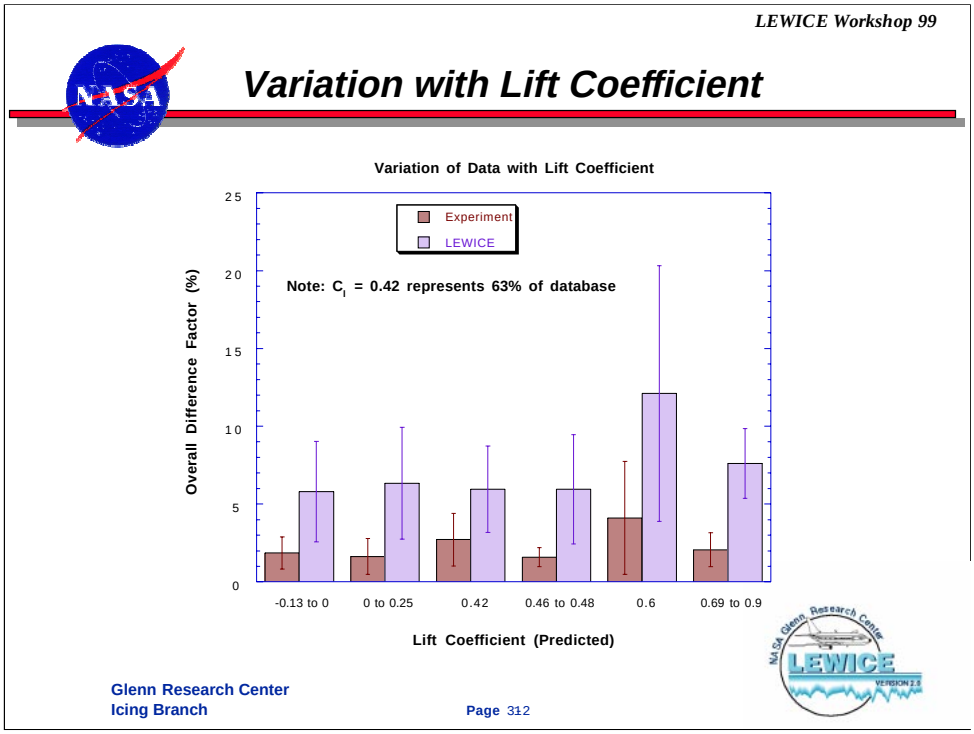
This chart shows the variation of the data with respect to Reynolds Number. This variable was chosen since the potential flow module was used in all of the LEWICE predictions which discounts viscous effects. Here, there is a significantly higher error in the 6.2-6.4 million range. Note the large breaks in between data ranges, which may signify that there is a physical meaning to this error. This error is not the expected trend with Reynolds Number. The following charts shed more light on the source of this error.



This chart shows the variation with respect to angle of attack. Here, there is a significantly higher error at the higher angles of attack than at lower angles of attack. It is worth noting that a majority of the data is at a 4˚ angle of attack, which means that other bins are under represented. Also note that the highest angle of attack was 7˚, so the airfoils are not near stall for the clean airfoil. Some cases may be approaching stall for the iced airfoils however. With respect to the previous chart, many of the data points in the 6.2-6.4 million Reynolds Number range were at a 6˚ angle of attack. It is believed that this factor contributes to the high error in that range on the previous chart.

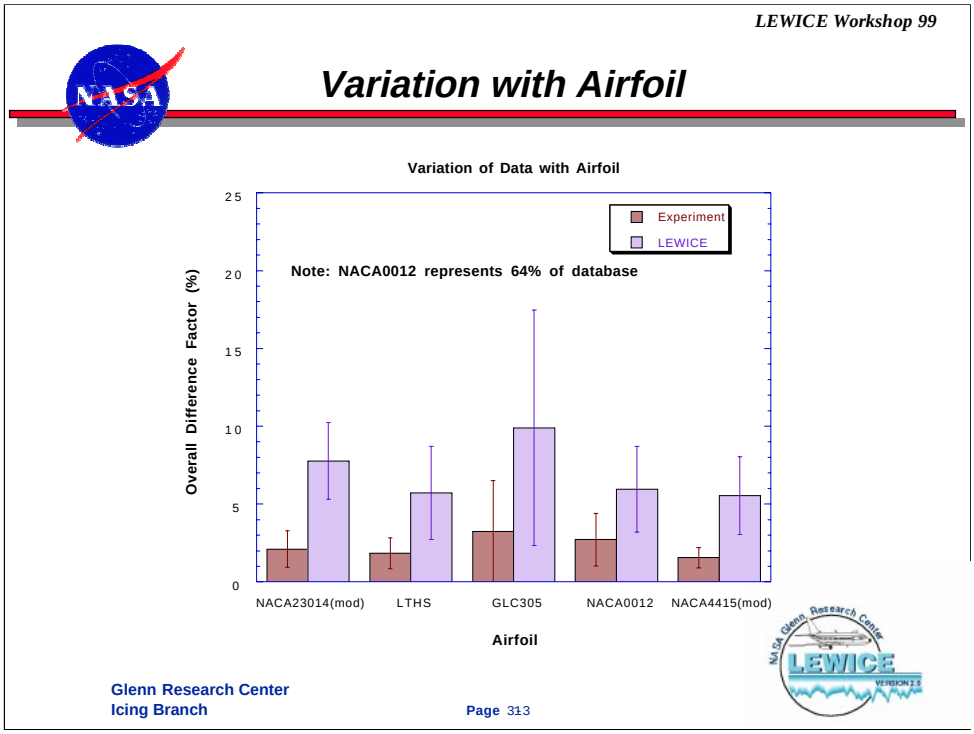
Conclusion: LEWICE 2.0 has higher errors in the upper angles of attack, perhaps indicating some stall or near stall behavior on the iced airfoils which is not being modeled.

Conclusion: The errors at high angles of attack contributes to the high error in the 6.2-6.4 million Reynolds Number region.



This chart shows a similar trend as the previous plot. It shows the errors as a function of the predicted lift coefficient on the clean airfoil. The lift coefficient shown is the one predicted by the potential flow solver in LEWICE. A lift coefficient of 0.42 is the predicted lift on a NACA0012 airfoil at 4° AOA and represents 63% of the database. A lift coefficient of 0.6 is the predicted lift coefficient of the GLC305 airfoil at 6° AOA. This chart shows that the error is higher with increasing lift coefficient, which is a further indication of possible stall or near stall conditions on some of those airfoils. Also note that the experimental error is higher in this region as well.

Conclusion: The error, both experimentally and from LEWICE, is higher for higher lift coefficients, especially the 0.6 lift bar data.



The final chart shows the error as a function of airfoil. Once again, it must be pointed out that the NACA0012 airfoil represents 64% of the database. This chart shows a similar result shown in the previous three charts. The relative error is higher for the GLC305 airfoil than for the other airfoils. Much of this data was taken at a 6˚ AOA at Reynolds Numbers in the 6.2-6.4 million range with a predicted lift coefficient of 0.6. The consistency of this error for these cases shows a strong indication of possible physical improvements which could be made in these cases.

Conclusion: The data for a GLC305 airfoil at 6˚ AOA, lift coefficient of 0.6, and Reynolds Number of 6.2-6.4 million consistently shows a higher error than the rest of the database.

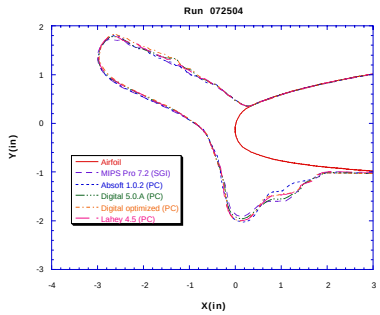
Validation Issues

- **Platform sensitivity**
- **Platforms where executable is available**
- **Source code availability**
- **Features not validated**
- **Validation on different platforms**
- **Validation if changes are made**
- **Forms for code requests**

The rest of this presentation will discuss other validation issues with LEWICE. The discussion will cover platform sensitivity and different ways of recompiling the source code, features of LEWICE 2.0 which were not validated and other issues.

Platform Sensitivity

- **Minor differences based on platform and/or compiler**
- **Validation requires separate tests for each platform**



This chart shows the representative difference in ice shape based on the platform and/or the compiler used. It also shows that simply changing the compiler flags can also have a small effect. All but one of the benchmark cases showed the small error exhibited in this plot. If LEWICE is ported to a platform it has not been tested on, the user should run the benchmark cases provided on the CD-ROM to establish the code validity on that platform.

Note: In the tests performed at NASA Glenn, the PC executable provided on the CD-ROM produced exactly the same results when run from different PCs. Included in this test were machines equipped with Windows 95 and 98, fast processors and slower machines as desktops as well as laptops.

Platform Availability

- **PCs**
  - **DOS/Win3.1/Win95/Win98**
  - **No known conflict with any specific brand**
- **Silicon Graphics**
- **Sun**
- **Macintosh**
  - **Tests incomplete**

The benchmark cases on the CD-ROM were run on several PCs using different operating systems, Silicon Graphics workstations and one Sun workstation. An executable could also be generated for the Macintosh platform, although the benchmark cases have not been run on that platform. The code is also expected to run on other systems, but the user would need to recompile the source code and run the benchmark cases to verify this hypothesis. The PC executable provided on the CD-ROM is expected to run on Windows/NT systems, but this could not be verified.

Source Code Availability

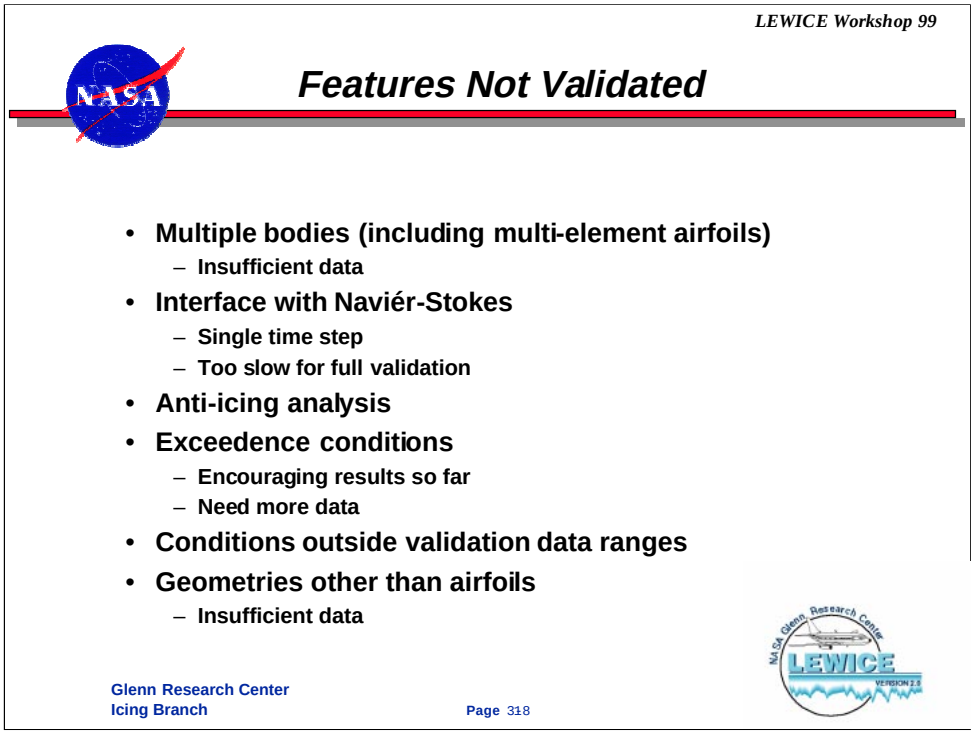
- **Source code for official 2.0 release freely available**
  - **Make separate request for tracking purposes**
- **Need to revalidate for each platform or compiler**
  - **Including different versions of compiler and flag settings**
- **Full validation required if code changes**
- **Source code modified by users cannot be supported by NASA**
- **Source code for interim NASA versions may not be available**
  - **Limited manpower for revalidation efforts**

Source code for LEWICE 2.0 is not included on the distribution CD-ROM. A separate request must be made for our tracking purposes. This was done to establish version control of the software and was not meant to restrict access to the source code.

If the source code is recompiled on another platform or with a different compiler or even different compiler flags, the user must run the 16 benchmark cases provided on the CD-ROM to establish that the cases run adequately for those settings.

If a developer makes changes to the LEWICE source code, the entire validation database (CR 208690) must be run with the modified code before NASA will consider the changes validated. In addition, the results must show quantitative improvement from version 2.0 to be considered an improvement. This statement holds true for versions created by NASA Glenn as well as those created by others.

As such, NASA reserves the right to not release interim version of LEWICE past version 2.0 since there is limited manpower available for revalidation efforts.



The list above shows the features of LEWICE 2.0 which could not be validated. **This does not mean that the code will not run for these cases nor that the results are necessarily inaccurate, but that validation was not performed.**

Multiple body simulations, which includes multi-element airfoils, could not be validated because publicly available data does not exist in sufficient quantities. **No ice accretion code, not matter how sophisticated the model, can be considered valid without such a database.**

The interface with grid-based codes (Euler as well as Naviér-Stokes) can not be validated as it is currently limited to single time step cases. Additionally, cases with enough grid resolution for complex ice accretion shapes can take up to full day (24 hrs) with the fastest available Naviér-Stokes codes for a single time step, rendering validation efforts too time-consuming at present.

The anti-icing feature was not compared with available data due to time constraints.

Exceedence conditions show very encouraging results to date. The available data is still somewhat limited however.

Conditions outside the validation database (CR 208690) or geometries dissimilar to those run (I.e. geometries other than airfoils) could not be validated because there is no publicly available database for the validation. LEWICE has been run on several non-airfoil configurations.

Validation Process

- **Validation procedure included with LEWICE 2.0 distribution**
- **Experimental test cases are needed to validate geometries dissimilar to those tested**
  - **Non-airfoils**
  - **Multi-element airfoils**
- **Procedure is readily adaptable to validate other modules if data becomes available**

The validation procedure for the benchmark cases or for full validation is provided on the CD-ROM. If the user wishes to validate LEWICE for conditions outside the validated region, they can run LEWICE in these regimes and compare the results against the experimental data. The user can also refer to the validation report, CR 208690.

The validation procedure is considered robust enough to be adapted toward validating the previous features listed if data becomes available for that validation.

Conclusions

- **Statistical variation of results shown**
- **Validation complete for version 2.0**
- **Nominal variation with different platforms**
- **More data is needed to validate some features**

The statistical variation of the LEWICE results shows higher error for a GLC305 airfoil at 6˚ AOA with a Reynolds Number in the 6.2-6.4 million range. Other conditions, including glaze ice cases and exceedence conditions do not show quantitatively higher errors.

There is nominal variation of results with different platforms, compilers or compiler flags necessitating some checks when the code is ported to a different machine.

Some features could not be validated due to lack of data. It is important to note that the code will run for those cases and we have confidence in the results. No code can be considered validated for regimes where comparisons cannot be made.

Discussion

- **Source code policy**
  - **Version control**
  - **Error tracking**
- **Validation process**
  - **Useful for certification?**

Are there suggested improvements to the validation techniques used?

Are there suggested improvements to the code release procedure?

What improvements can be made to ensure usefulness for certification efforts?